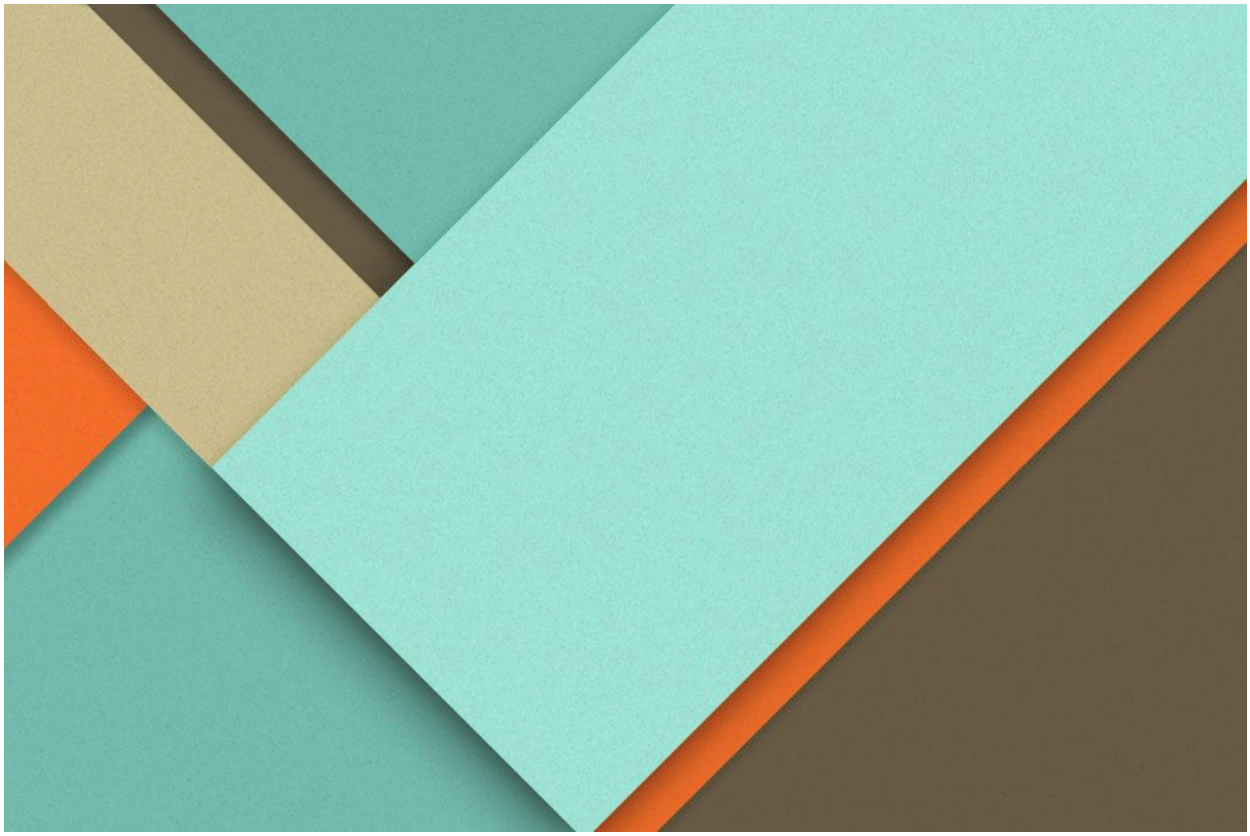

Water Supply Automation

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Automation Project for domestic space
Mechatronics Engineering Project

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Overview

Introduction

Residential water management in many regions remains a fragmented process, characterized by manual monitoring and open-loop control. Homeowners typically rely on visual inspection or mechanical float switches to manage storage levels and pump operations. However, these traditional methods are prone to human error and mechanical fatigue, often resulting in catastrophic pump failure due to dry-running or significant water wastage through overflows.

This project addresses these inefficiencies by developing an **intelligent, distributed control system** designed for residential water resource management. By transitioning from a manual approach to a **semi-automated, closed-loop architecture**, the system integrates embedded controllers with real-time sensor networks. The primary focus of this implementation is to optimize operational efficiency, provide robust protection for electromechanical hardware, and enhance overall user safety through autonomous fault detection.

The current architecture serves as a Functional Prototype. The next iteration involves migrating the control logic to an **Industrial PLC (e.g., Siemens S7-1200)**. This would replace the 433MHz link with **RS-485/Modbus** for electrical noise immunity and utilize industrial **24V DC logic** for integration with standard motor control centers (MCCs)

System Architecture

The infrastructure consists of a multi-stage storage configuration:

- **Primary Reservoir (Basement):** Fed by the municipal supply via a metered inlet.
- **Secondary Overhead Tank (Roof):** Responsible for pressurized domestic distribution, fed by a lift pump connected to the basement reservoir.

Problem Statement & Engineering Challenges

The legacy manual system presented significant operational risks and inefficiencies:

- **Human-Dependent Operational Constraint:** The legacy system required manual intervention at irregular intervals, specifically during late-night municipal supply windows. This created a high "Human-in-the-loop" dependency, where manual

monitoring was required to prevent tank overflows while simultaneously ensuring the pump was deactivated immediately upon reservoir depletion to avoid dry-run cavitation.

- **Pump Reliability (Dry Run Risk):** The lack of real-time feedback from the primary reservoir led to "dry-run" conditions, causing cavitation and thermal damage to the pump motor when municipal supply was interrupted.
- **Safety & Accessibility Hazards:** Manual level-checking required accessing high-risk zones (heavy steel hatchets over basement openings), presenting significant safety liabilities for residents.
- **Lack of Telemetry:** No centralized interface existed to monitor system status, leading to water shortages and inefficient usage patterns.

Assumptions & Constraints

Assumptions

- **Constant Volumetric Geometry:** It is assumed that both the Primary Reservoir and Secondary Tank are fixed parameters where the vertical distance measured by the ultrasonic sensor correlates linearly to water volume.
- **Constant Pump Head & Flow Rate:** We assume the pump operates at a consistent flow rate given a stable voltage. This justifies using **Time-Based Safety Cutoffs** as a secondary fail-safe to the current sensor.
- **Non-Turbulent Surface Conditions:** It is assumed that the inflow rate into the tanks does not create enough surface agitation to cause "noise" or false readings in the JSN-SR04T ultrasonic data.
- **Power Grid Recovery Logic:** We assume that upon power restoration after a "load-shedding" event, the system should enter a **Safe-Boot state** (60-second delay) rather than immediately triggering high-current inductive loads (the pump).

Constraints

- **Legal/Regulatory (Non-Invasive Sensing):** The municipal supply line is metered and protected by local utility laws. The system must be strictly "Post-Meter." No modification or tampering with the municipal inlet pipe is permitted, which precludes the use of inline flow meters at the source.
- **Physical/Structural (Fixed Plumbing):** The building's existing plumbing is encased in concrete or finished walls. Re-plumbing for flow sensors or internal float switches

is prohibited due to high labor costs and structural risk. The solution must be **Retrofit-Ready**, utilizing external or non-contact sensing (Ultrasonic).

- **Economic (Budgetary Cap):** As a student/independent project, the hardware budget is strictly limited. This dictated the use of 433MHz RF modules (cost-effective) over LoRa, and the choice of ESP-series microcontrollers over industrial PLCs.
- **Environmental (Power Instability):** The operating environment is subject to frequent and unpredictable power outages. The system must include non-volatile memory (EEPROM/LittleFS) to "remember" its state (e.g., *was the pump on before the power cut?*) to prevent logic loops upon reboot.
- **RF Interference:** The communication must pass through reinforced concrete floors. This limits the data packet size and requires a high-redundancy "Burst-and-Ack" protocol to overcome signal attenuation.

Objectives

Process Optimization & Problem Mitigation

To engineer a **closed-loop control system** that eliminates manual oversight in municipal water acquisition. The system will automate the transfer of water between storage stages while maintaining resource equilibrium.

Embedded Systems & Multi-Modal Sensing

To execute the integration of a distributed sensor network with ESP-series microcontrollers for high-fidelity state estimation. This involves implementing non-contact ultrasonic ranging (JSN-SR04T) for precise volumetric measurement and hall-effect current sensing (ACS712) for real-time load monitoring. By correlating distance data with electrical current profiles, the system achieves a 'multi-modal' safety check, distinguishing between successful water displacement and air-suction/cavitation events.

Industrial Design & HMI Development

To design a **cost-optimized Human-Machine Interface (HMI)** featuring a custom 3D-printed enclosure. The HMI will facilitate real-time telemetry (LCD data visualization) and provide physical control overrides to enhance operator situational awareness.

Communication & System Topology

To establish a robust data link across multi-story structural barriers, the system implements a custom **Water Automation Communication Protocol (WACP) v4.0**. This protocol utilizes a **Master-Slave architecture** over 433MHz FSK radio, employing a **Burst-and-Ack strategy** to ensure message delivery through reinforced concrete without frequency jamming. Reliability is further reinforced by a **multi-tier watchdog system** and **staggered heartbeat monitors** that automatically terminate pump operations if the data link is severed for more than three minutes.

Fault Tolerance & Fail-Safe Logic

To implement a **fail-safe architectural design**, incorporating software watchdogs and hardware interlocks. The system will prioritize hardware longevity by enforcing mandatory "Dry-Run" protection and providing emergency manual bypass modes.

Logic Schematic

Operational Thresholds & Setpoints

To ensure system longevity and prevent critical "stock-out" events, the system operates within defined **Operational Envelopes**. Rather than utilizing absolute zero (0%), the system maintains a safety buffer to prevent sediment ingestion and pump cavitation.

Storage Unit	Low-Level Setpoint (LL)	High-Level Setpoint (LH)	Purpose
Reservoir (C1)	25%	80%	Buffer for municipal fluctuations.
Overhead Tank (C2)	25%	80%	Buffer for peak domestic demand.

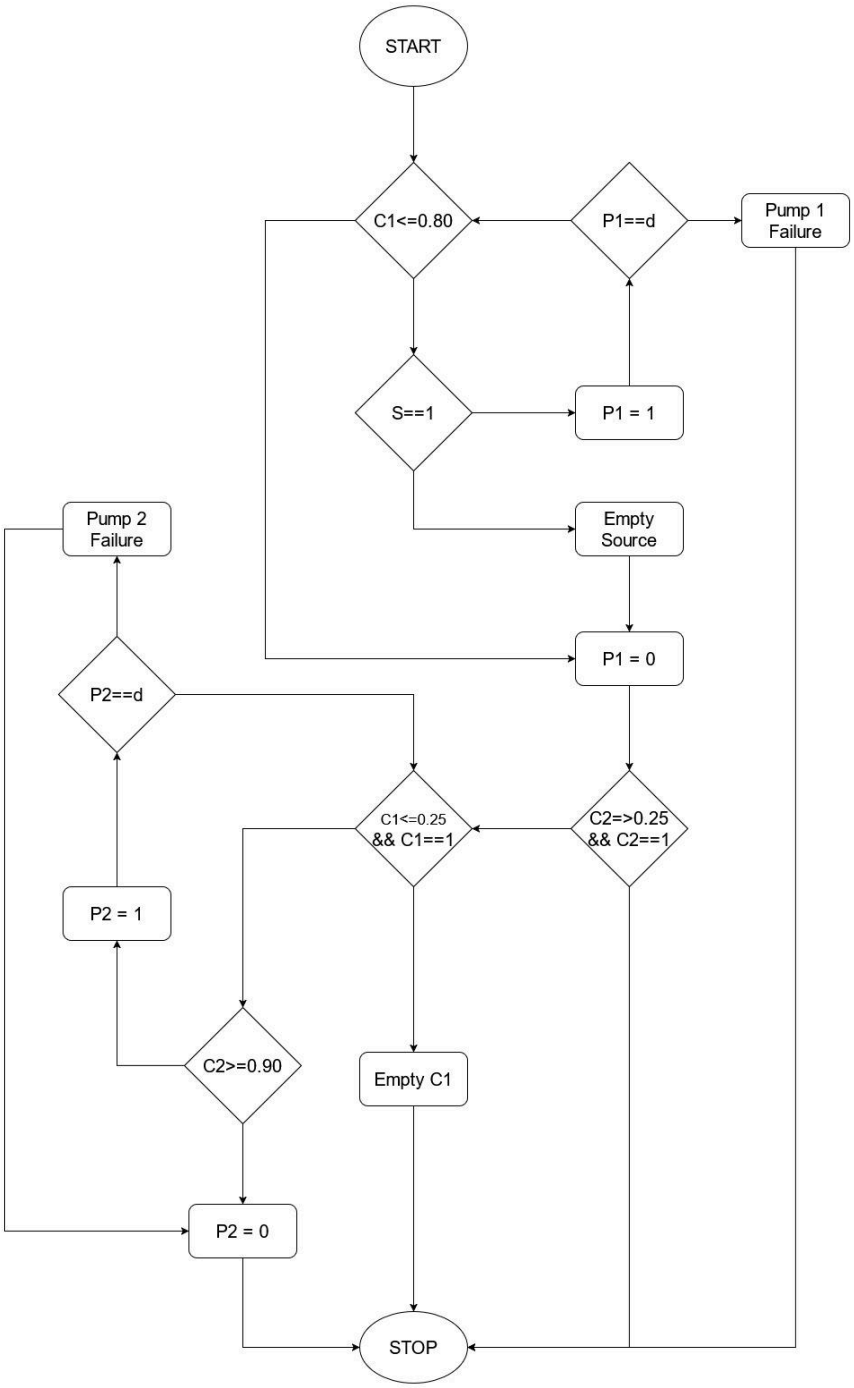

fig.1) Flow Chart

System State Architecture

The logical behavior of the BareBone OS is defined by three distinct Finite State Machines (FSMs) operating in parallel across the network. The following sections detail the operational flow for the Sump Station (SC1), Pump Station (SC2), and the Master Control Unit.

Overhead Station (SC1) – The Event Generator

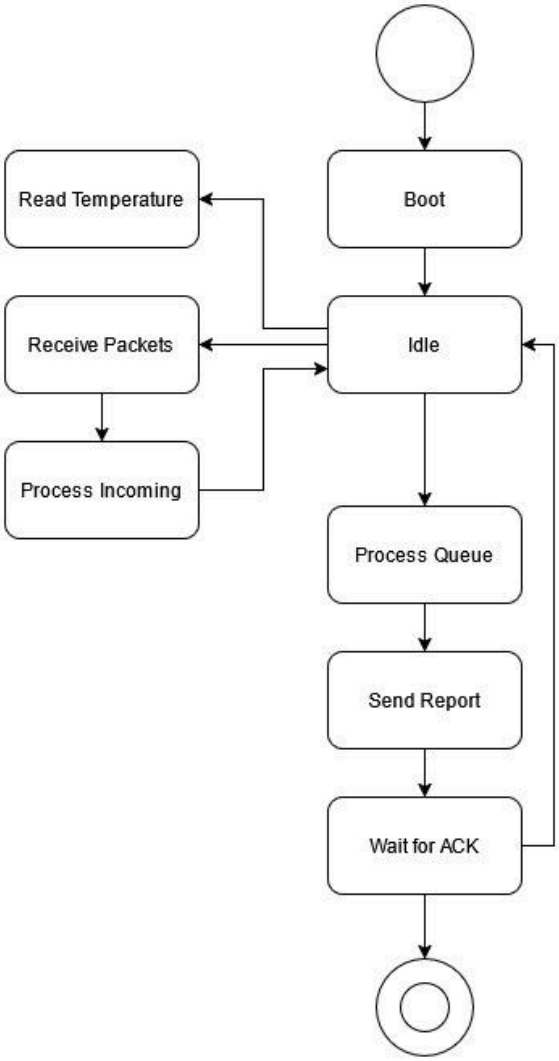

Fig 2) SC1 State Transition Diagram

SC1 States

1. **BOOT** — Initialize pins, begin serial and HC12, begin DHT11, take baseline ultrasonic reading
2. **IDLE** — Main loop, waiting for triggers
3. **READ_TEMPERATURE** — DHT11 reading every 10 minutes, only if channel clear
4. **READ_LEVEL** — JSN-SR04T fires, calculates distance using current temperature, returns percentage
5. **RECEIVE_PACKET** — 6 bytes available on HC12, read and validate preamble, destination, checksum
6. **PROCESS_INCOMING** — Route valid packet, handle CMD_PING or CMD_ACK
7. **SEND_REPORT** — Build and push level report to queue
8. **WAIT_FOR_ACK** — Wait up to 630ms, retry up to 3 times, discard gracefully on failure

The Overhead Station (SC1) operates on a **Wake-on-Event** architecture designed to minimize radio traffic and ensure high-precision telemetry. The system logic follows these primary paths:

External Trigger (Polling & Synchronization) When a "PING" or "GET_LEVEL" packet is received from the Master, SC1 transitions to the **PACKET RECEIVED** state. It immediately generates a health report. To prevent collisions, SC1 utilizes a **1300ms Backoff Timer** before attempting transmission, ensuring it does not overlap with SC2 or Master broadcasts.

Internal Trigger (Environment & Level Change)

- **Precision Level Sensing:** SC1 utilizes a **JSN-SR04T** ultrasonic sensor. Every 10 minutes (provided the HC-12 channel is clear), a **DHT11 sensor** triggers a temperature reading. The speed of sound is dynamically recalculated using the formula: $v=331.3+(0.606\times temp)$. This ensures the 80% **TANK_HIGH** threshold is accurate regardless of ambient weather conditions.
- **Asynchronous Event Logic:** If a significant level shift is detected, SC1 bypasses standard polling and pushes a report to the **8-slot Priority Queue**.

Priority Transmission & Reliability The "Transmission Queue" is now a **Priority-Based Stack**. Packets are categorized as follows:

- **Priority 1/2:** Critical Faults or State Changes.
- **Priority 3:** Normal Level Reports.

- **Priority 4:** ACKs. The **WAITING FOR ACK** state remains the final safeguard, confirming the Master has logged the data before SC1 clears that specific slot from the queue.

6.2 Sump Station (SC2) – The WorkHorse

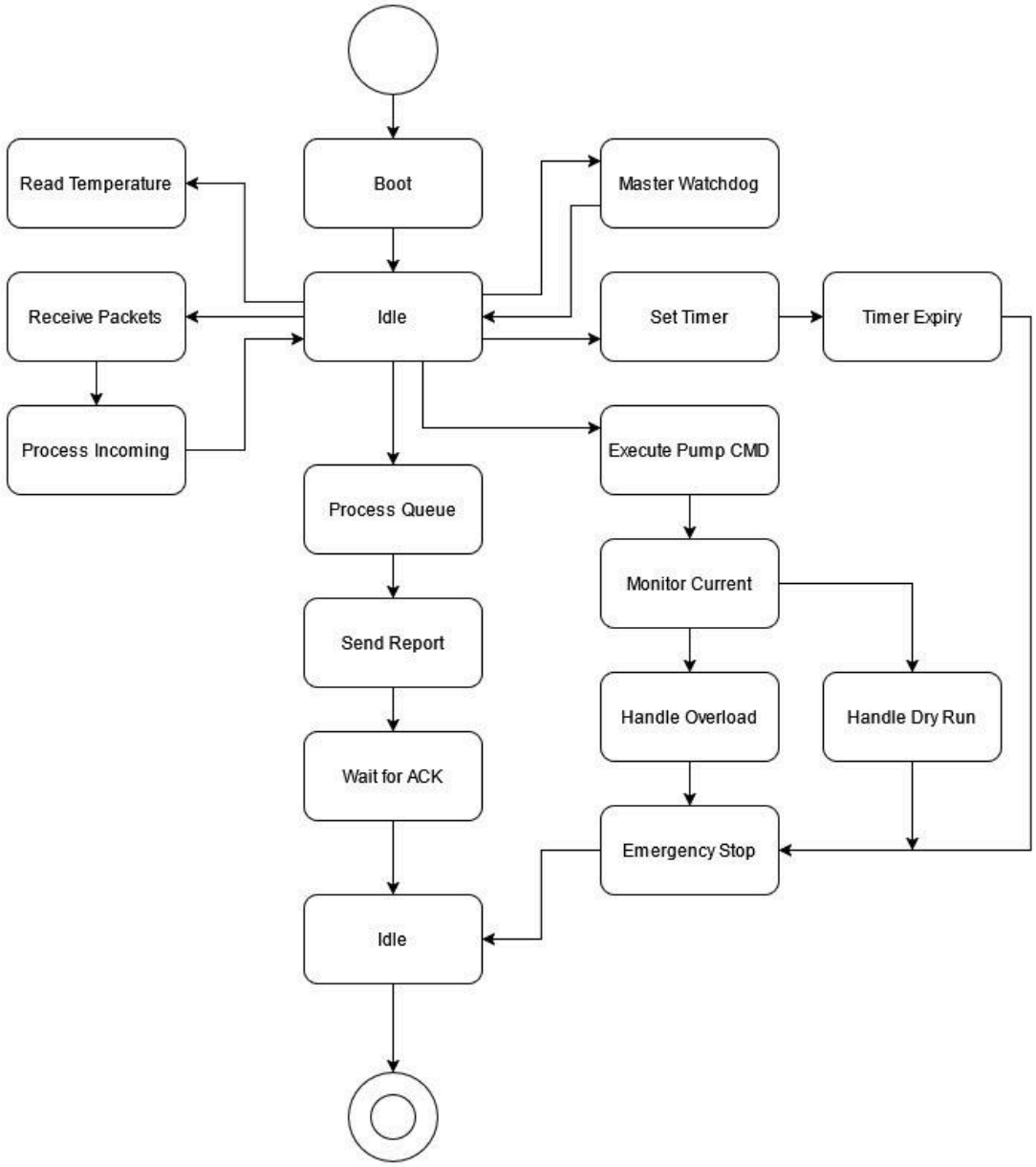

Fig 3) SC2 State Transition Diagram

SC2 States

1. **BOOT** — Initialize pins, relays OFF, begin serial and HC12, begin DHT11, take baseline ultrasonic reading
2. **IDLE** — Main loop, waiting for triggers
3. **READ_TEMPERATURE** — DHT11 reading every 10 minutes, only if channel clear
4. **READ_LEVEL** — JSN-SR04T fires, calculates distance using current temperature, returns percentage
5. **RECEIVE_PACKET** — 6 bytes available on HC12, read and validate preamble, destination, checksum
6. **PROCESS_INCOMING** — Route valid packet, handle CMD_PING, CMD_PUMP, CMD_TIMER, CMD_ACK
7. **EXECUTE_PUMP_COMMAND** — Activate or deactivate relay based on command value, respect locked and fault flags
8. **SET_TIMER** — Store timer end timestamp, set timerActive flag
9. **TIMER_EXPIRY** — Timer end reached, P2 OFF, timerActive cleared, report pump state
10. **MONITOR_CURRENT** — Read ACS712 via multiplexer for active pump, check against dry run and overload thresholds
11. **HANDLE_DRY_RUN** — Pump OFF, increment retry count, wait 10 seconds, retry up to 3 times before escalating
12. **HANDLE_OVERLOAD** — Both pumps OFF immediately, no retries, escalate to emergency stop
13. **EMERGENCY_STOP** — Cancel timer, kill both relays, push CMD_FAULT and CMD_AMP to queue at critical priority
14. **SEND_REPORT** — Build and push level report to queue
15. **PROCESS_QUEUE** — Transmit highest priority packet, handle retries on 500ms backoff
16. **MASTER_WATCHDOG** — Monitor time since last Master packet, freeze P2 after 2 minutes silence, report on reconnect

The Sump Station (SC2) acts as the system's **"Intelligence & Power Hub,"** managing high-power actuators and multi-channel safety sensors. The architecture is governed by an advanced feedback and protection loop:

Multi-Channel Hardware Abstraction

- **Dual Current Monitoring:** SC2 utilizes a multiplexer on **Pin D7** to cycle between two ACS712 sensors. When **D7 is LOW**, it monitors Pump 1 (P1); when **D7 is HIGH**, it monitors Pump 2 (P2).

- **Environment-Aware Sensing:** To ensure the **80% TANK_HIGH** limit is precise, SC2 uses a **DHT11** to calibrate the speed of sound for the JSN-SR04T ultrasonic sensor every 10 minutes.

The "Double-Tap" Fault Protocol The **SENSOR MONITORING** block now performs a more sophisticated analysis. If a Dry Run or Overload is detected:

- **Packet 1 (CMD_FAULT):** Immediate broadcast of the fault type.
- **Packet 2 (CMD_AMP):** Follow-up packet containing the exact amperage (Encoded as VAL × 10).
- **Priority:** These are pushed to **Priority 1 (CRITICAL)** in the 8-slot queue, bypassing all level reports.

The "Locked" Safety State The system no longer just enters "Emergency Stop." It now implements a **`isLocked` flag**.

- **The Logic:** Once a fault occurs, the pump is logically locked. Even if the fault condition clears, the **Auto-Logic cannot restart the pump**.
- **Reset:** A manual intervention from the Master's pump screen is the only way to clear the `isLocked` state.

Communication Watchdog (Fail-Safe) SC2 monitors a **DISCONNECT_PERIOD (120,000ms)**. If the Master is silent for 2 minutes, the **Disconnect Watchdog** triggers:

- **Action:** Pump 2 (P2) is immediately frozen to prevent uncontrolled operation during a comms blackout.
- **Recovery:** Upon reconnection, SC2 automatically synchronizes its current P2 state back to the Master.

Collision Management To ensure the Master stays in control, SC2 uses a **383ms Backoff Timer**. This allows it to transmit faster than SC1 (521ms) but slower than the Master (250ms), prioritizing reservoir/pump data over overhead tank data.

6.3 Master Control Unit – The Central Arbiter

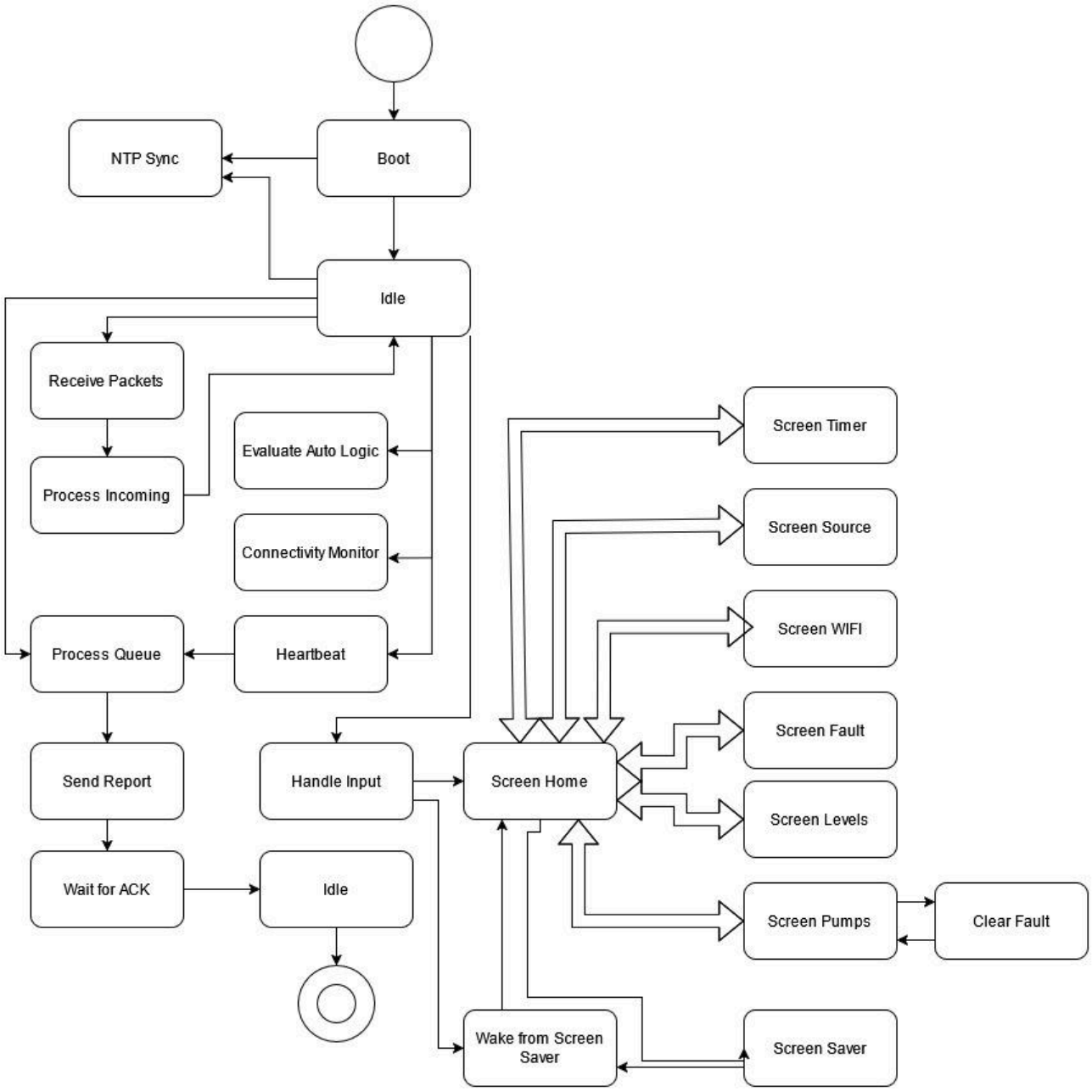

Fig 4) Master Control State Transition Diagram

Control States

1. **BOOT** — Initialize pins, begin HC12, init display, draw static frame
2. **NTP_SYNC** — Connect WiFi, get time, disconnect, schedule next sync at 7AM
3. **IDLE** — Main loop, all monitors running simultaneously
4. **SCREENSAVER** — Activated after 60 seconds inactivity, display screensaver images, any button wakes
5. **WAKE_FROM_SCREENSAVER** — Restore static frame, redraw previous screen
6. **UPDATE_CLOCK** — Redraw time string only if time has changed, called every loop
7. **HEARTBEAT** — Send ping to SC1 at 1 second mark, SC2 at 31 second mark, reset on 31 second cycle
8. **RECEIVE_PACKET** — 6 bytes available on HC12, validate preamble, destination, checksum
9. **PROCESS_INCOMING** — Route valid packet, update system state, update display
10. **HANDLE_INPUT** — Read analog A0, route to Toggle, Select or Back handler
11. **SCREEN_HOME** — Menu navigation, cursor cycles through 5 items on toggle
12. **SCREEN_PUMPS** — Pump selection and ON/OFF action, heart cursor, lightning indicators
13. **SCREEN_LEVELS** — Static tank images, live C1 and C2 percentage display
14. **SCREEN_TIMER** — Preset selection, countdown display, cross shown when timer active
15. **SCREEN_SOURCE** — YES/NO source toggle, disabled when timer active or C2 full
16. **SCREEN_WIFI** — SSID and IP display, connection diagnostic with hourglass animation
17. **SCREEN_FAULT** — Fault display with pump name, reason and amperage, buzzer alarm, locked until SELECT
18. **EVALUATE_AUTO_LOGIC** — P1 logic driven by source flag and C1 level, P2 logic driven by C2 demand and C1 availability, respects locked flags
19. **CONNECTIVITY_MONITOR** — 180 second timeout for SC1 and SC2, kill pumps if SC2 goes silent
20. **PROCESS_QUEUE** — Transmit highest priority packet, handle retries on 250ms backoff
21. **CLEAR_FAULT** — Lock faulted pump, clear fault code, restore previous screen

The Master Unit acts as the **High-Speed System Coordinator** and **HMI (Human-Machine Interface)**. It manages the flow of the BareBone OS protocol while maintaining a strict safety barrier between user commands and hardware execution.

Instant-Access User Interface (Manual Input) Unlike the satellite nodes, the Master has **Zero Boot Delay**. The UI is available immediately upon power-up.

- **Extended Scheduling:** The timer system has been expanded from 3 presets to **9 steps** (30-minute increments up to 4 hours 30 minutes), providing finer control over tank filling cycles.
- **The Safety Gate:** If a pump is in a **Locked (`isLocked = true`)** state due to a previous fault, the Master will ignore all "Auto" logic requests. The user MUST navigate to the "Pumps Screen" and manually trigger to reset the flag.

Priority-Driven Network Orchestration (Listen Radio) The Master utilizes a high-speed **9600bps baud rate** and the most aggressive timing in the system with a **250ms Backoff Timer**. Communication is managed via an **8-slot Priority Queue**:

1. **Priority 1:** `CMD_PUMP` (Instant action)
2. **Priority 2:** `CMD_TIMER` (Scheduling)
3. **Priority 3:** `CMD_PING` (Health checks)
4. **Priority 4:** `CMD_ACK` (Confirmations)

Advanced Telemetry & Fault Processing The Master now handles a "Double-Tap" data flow from SC2:

- **Packet Reception:** When a fault occurs, the Master expects two consecutive packets: **CMD_FAULT** followed by **CMD_AMP_Px**.
- **Data Decoding:** The Master receives the amperage as an integer (VAL × 10) to save bandwidth and decodes it back to a float (VAL / 10.0) for the UI display.
- **ACK Logic:** The Master holds the Acknowledge (ACK) signal until both the Fault and the specific Amperage data are successfully logged into the local registers.

Heartbeat & Watchdog Maintenance The HEARTBEAT state ensures the **Disconnect Watchdog** on SC2 remains satisfied. By maintaining a constant polling cycle, the Master prevents SC2 from "freezing" P2. If the Master fails to send a signal for 120 seconds, it assumes a system-wide comms failure and prepares for a re-sync upon reconnection.

Control Algorithm & Logic Flow

The system employs a **Calibrated Asynchronous Monitoring Loop**. Automation is governed by a safety-first hierarchy where hardware protection (Amperage/Watchdog) overrides all operational commands.

Sensor Calibration Layer (The Foundation)

Before any logic execution, the system validates environment data:

- **Temperature Compensation:** SC1 and SC2 read the **DHT11** every 10 minutes to update the speed-of-sound constant. This ensures the **80% TANK_HIGH** threshold is physically accurate.
- **Current Sensing Prep:** SC2 toggles **Pin D7** to select the correct ACS712 channel (**LOW for P1**, **HIGH for P2**) before sampling amperage.

Primary Logic (Municipal Supply → C1)

The system uses the expanded **9-step Timer (30m to 4.5h)** or a manual HMI trigger to initiate P1:

- **Safety Gate:** `IF (P1_isLocked == TRUE) → REJECT START`.
- **Execution:** Activate P1 + Set D7 LOW.
- **Fault Logic:** If P1 Amperage (decoded as VAL / 10.0) indicates a Dry Run or Overload, SC2 sends a **Double-Tap report (CMD_FAULT + CMD_AMP)**, sets `P1_isLocked = TRUE`, and shuts down.

Secondary Logic (C1 → C2 Transfer)

The transfer logic follows a **Watchdog-Protected Interlock**:

- **Automation Logic:** * **START P2 IF:** `Level(C2) < 25%` AND `Level(C1) > 25%` AND `P2_isLocked == FALSE`.
    - **STOP P2 IF:** `Level(C2) ≥ 80%` OR `Level(C1) ≤ 20%`.
- **The Watchdog Interrupt:** If the **120,000ms Disconnect Watchdog** expires (no signal from Master), P2 is immediately frozen in its last safe state (usually OFF) to prevent reservoir depletion during a communication blackout.

Fault Recovery & Priority

- **Manual Override:** Once a pump enters the `isLocked` state, it cannot be restarted by the timer or auto-logic. A **Manual Unlock Command** must be sent from the Master's Pump Screen.

- **Traffic Priority:** Any change in pump state or fault detection generates a **Priority 1 or 2 packet**, which jumps to the front of the 8-slot queue, ensuring the Master reacts within 250ms of the event.

Semi-Automated Input & Feedback Loop

Due to the non-deterministic nature of municipal water schedules, the system utilizes a **High-Priority Manual Interface** on the Master Unit. This allows the operator to synchronize automation with supply availability while the system provides real-time safety telemetry.

Synchronized "Supply Windows" The operator no longer simply toggles a switch. Instead, they select from **9 Timer Presets** (ranging from 30 minutes to 4 hours 30 minutes in 30-minute steps). This allows the user to match the pumping cycle to the known duration of the municipal supply.

- **Immediate Availability:** Because the Master Unit has **no boot delay**, the operator can initiate a cycle the moment municipal water arrives, even if the satellite stations (SC1/SC2) are still finishing their 60-second stabilization.

The Safety Feedback Loop (Verification) Once a "Supply Window" is initiated, the system performs an active verification of the operator's input:

- **Active Monitoring:** As P1 begins drawing from the municipal line, the SC2 station monitors the **ACS712 (via D7 LOW)**.
- **Dry-Line Detection:** If the operator acknowledges supply but the line is actually dry, the system detects a low-amperage fault within seconds.
- **Lock-Out Response:** The system immediately sets the **`isLocked` flag** and sends the "Double-Tap" fault report to the Master. The HMI then shifts from "Pumping" to a **"FAULT: P1 LOCKED"** state.

Manual Intervention & Reset The feedback loop requires a "Human-Clear" for high-stakes errors:

- **Non-Auto Recovery:** If a fault occurs during a manual cycle, the timer is cleared, and the pump is locked. The auto-logic cannot "retry."
- **Manual Unlock:** The operator must manually interact with the Master's pump screen to **Unlock** the system. This ensures the operator has physically checked the municipal line or pump intake before the system allows another attempt.

Command Priority Every manual input (Start, Stop, Timer Change, Unlock) is pushed into the **Priority 1 (CRITICAL)** slot of the Master’s 8-slot queue. This ensures that the user’s

command is transmitted within **250ms**, providing the "snappy" feel of a hard-wired switch despite the wireless architecture.

Fault Diagnostics & Error Handling

The system utilizes an **Advanced Telemetry & Safety Layer** designed to protect expensive pump hardware and prevent water wastage. Diagnostics are no longer just "binary" (on/off); they are now **quantitative**, providing real-time amperage data for every failure.

Expanded Failure Modes (The "Six-Point" Check)

The system differentiates between specific electrical and hydraulic failures using the ACS712 sensors:

- **Dry Run (P1_DRY / P2_DRY):** Detected when Amperage < *Lower Threshold*. Indicates the municipal line is dry or the reservoir (C1) is empty.
- **Overload (P1_OVER / P2_OVER):** Detected when Amperage > *Upper Threshold*. Indicates a jammed impeller, internal short, or motor fatigue.
- **Hardware/Open Circuit (P1_FAIL / P2_FAIL):** Detected when the relay is triggered but Amperage is 0.0A. Indicates a blown fuse, tripped breaker, or broken wiring.

The "Double-Tap" Reporting Protocol

To save bandwidth while providing maximum detail, the system uses a two-packet high-priority burst:

1. **Packet A (CMD_FAULT):** Immediate notification of the fault type (Priority 1).
2. **Packet B (CMD_AMP_Px):** Follow-up packet containing the exact current reading at the moment of failure (Encoded as VAL × 10). The Master unit decodes this (VAL / 10.0) and displays the exact trip-current on the screen (e.g., *"FAULT: P1 OVERLOAD @ 8.4A"*).

The "Locked" Safety State

Every fault listed above triggers the **`isLocked` flag** for the affected pump.

- **Logical Isolation:** Once a pump is locked, the Master removes it from the automation pool.
- **Manual Clearance:** To prevent "fault-looping" (where a system keeps trying to restart into a short circuit), the error **must be manually acknowledged and UNLOCKED** from the HMI.

Communication Watchdog (System Vitality)

A new "Silent-Failure" diagnostic has been added:

- **DISCONNECT_PERIOD (90s):** If the Master fails to "Ping" the system for 2 minutes, SC2 identifies a **Comms Timeout**.
- **Fail-Safe Action:** SC2 freezes Pump 2 (P2) to prevent the reservoir from being drained into the roof while the system is "blind."
- **Auto-Recovery:** Once the Master reconnects, the priority queue synchronizes the current pump states to ensure the UI matches the hardware.

Array Holders

The system employs several structured arrays across its stations to manage state, communication, and user interface data in a clean and maintainable manner. Rather than relying on scattered individual variables, related data is grouped into arrays and structures to allow indexed access, reduce code duplication, and improve overall readability. These arrays are declared globally and persist throughout the lifetime of the program, serving as the single source of truth for their respective domains.

Pump State Array The `pumps[]` array, present on both the Control and SC2 stations, is a two-element array of the `PumpState` structure. Each element corresponds to one pump — index 0 for the Overhead pump and index 1 for the Basement pump. The structure encapsulates all relevant per-pump information including its current on/off state, fault status, lockout flag, fault code, dry run retry count, and a human-readable status label. Centralizing pump data in this manner ensures that both the auto logic and the display functions always reference a single consistent source.

Packet Queue Array The `packetQueue[]` array, present on both the Control and SC2 stations, is an 8-slot array of the `QueuedPacket` structure. Each entry stores the raw 6-byte packet data alongside its assigned priority level, current retry count, and acknowledgement status. This array forms the backbone of the priority-based transmission system, ensuring that critical fault packets are always dispatched ahead of routine reports and acknowledgements.

Timer Preset Arrays The `timerPresets[]` and `timerLabels[]` arrays are present exclusively on the Control station and together define the nine available timer presets. The `timerPresets[]` array stores the duration of each preset in minutes ranging from 30 to 270 in 30-minute increments, while `timerLabels[]` stores the corresponding human-readable strings displayed on the timer screen such as "1 hr", "2 hr 30 min" and so on. Both arrays are indexed together using a single `timerIndex` variable that advances on each toggle press.

Packet Buffer Arrays The `inPacket[]` and `outPacket[]` arrays are present on all three stations and each hold exactly 6 bytes corresponding to the fixed WACP v4.0 packet structure. The `outPacket[]` buffer is populated by the packet builder before each transmission, while `inPacket[]` receives and holds incoming bytes pending preamble validation, destination verification, and checksum confirmation. These buffers are intentionally kept minimal at 6 bytes each to reduce RAM overhead on the memory-constrained NodeMCU platform.

System Architecture

Microcontroller Unit (MCU): NodeMCU (ESP8266)

The NodeMCU was selected for its high performance-to-cost ratio and integrated WiFi stack, which facilitates future IoT integration without additional hardware. Its small form factor allows for a compact PCB footprint within 3D-printed enclosures while providing sufficient GPIO pins for I2C and SPI peripheral interfacing. Furthermore, the 3.3V logic level is compatible with the low-power sensor array selected for this project.

While PLCs offer superior electrical noise immunity and industrial-grade housing, they were rejected due to **prohibitive unit costs** and **excessive hardware overhead**. The prototype intentionally uses ESP-class controllers for rapid iteration and cost efficiency, with the control logic structured to allow direct migration to IEC 61131-3 PLC environments. Additionally, the proprietary nature of PLC software would limit the custom algorithmic control required for the project's specific safety watchdogs.

Power Distribution: Isolated SMPS & Protection

To ensure user safety and circuit longevity, an **Isolated Switch-Mode Power Supply (SMPS)** is utilized to convert 220V AC to a regulated 5V DC rail.

Safety Note: While the prototype currently utilizes a **T5A 250V Slow-Blow Fuse** due to existing inventory, the final design specification mandates a **T1A 250V fuse** to more accurately match the lower current draw of the logic circuitry and provide tighter overcurrent protection.

Level Sensing: JSN-SR04T Waterproof Transducer

Unlike mechanical float switches, which are prone to fouling and corrosion, the **JSN-SR04T** uses non-contact ultrasonic ranging. The IP67-rated waterproof probe is essential for high-humidity environments inside the reservoir, ensuring reliability against condensation and water splashes.

Diagnostic Sensing: ACS712-30A Hall-Effect Current Sensor

A current-based diagnostic approach was chosen over mechanical flow sensors to minimize plumbing modifications and reduce costs. By monitoring the current signature of the pump, the system can detect:

- **Under-load:** Indicating dry-running (low current).
- **Over-load:** Indicating mechanical blockage or stall (high current).

Communication: HC - 12433 SI4463

Due to strong signal attenuation caused by reinforced concrete structures and multi-floor barriers in the deployment environment, a robust sub-GHz wireless communication solution was required. The **HC-12 (433 MHz SI4463-based transceiver)** was selected to replace the earlier SRX882 ASK module.

Compared to basic ASK transmitter–receiver modules, the HC-12 provides:

- **Higher reliability** – built-in packet handling and error checking reduces packet loss.
- **Better noise immunity** – frequency shift keying (FSK) modulation is more resistant to interference than ASK/OOK.
- **Longer communication range** – up to ~1 km line-of-sight with higher transmit power and receiver sensitivity.
- **Stronger wall penetration** – 433 MHz sub-GHz signals perform better through concrete and floors.
- **Bidirectional communication** – true transceiver operation simplifies acknowledgements and feedback.
- **Simple UART interface** – easy integration with microcontrollers without complex RF tuning.
- **Stable data transmission** – consistent communication suitable for automation and control systems.

These features make the HC-12 more suitable for the water automation system, ensuring dependable communication between nodes in environments where Wi-Fi and basic RF modules experience significant signal loss.

Actuation Module: 30A Opto-Isolated Relay Module

Domestic water pumps have a high **inductive inrush current** (initial surge) that can weld the contacts of standard 10A hobbyist relays. A 30A-rated relay provides a necessary factor of safety. The **optocoupler** ensures galvanic isolation, preventing high-voltage spikes from back-feeding into the NodeMCU's sensitive logic.

Signal Conditioning: Bi-directional Logic Level Converters (BSS138 or Resistor Voltage Dividers).

The NodeMCU operates on **3.3V CMOS logic**, whereas the JSN-SR04T and ACS712 sensors output **5V TTL signals**. Integrating level shifters prevents over-voltage stress on the ESP8266 GPIO pins, ensuring the long-term reliability of the silicon.

Temperature & Humidity Module: DHT-11 Sensor

Since ultrasonic distance measurement depends on the speed of sound in air, which varies with environmental temperature and humidity, a temperature–humidity sensing module was integrated to compensate for these variations. By dynamically adjusting the speed of sound used in distance calculations, measurement accuracy and system reliability are improved under changing atmospheric conditions.

$v \approx 331.3 + (0.606 \times T)$

Human-Machine Interface (HMI) Module

To facilitate real-time monitoring and manual intervention, a centralized HMI node is implemented with the following peripherals:

Visual Interface: 20x4 I2C LCD

A 20x4 character display was chosen over the standard 16x2 to allow simultaneous telemetry of the Top Tank Level, Reservoir Status, Pump Current, and System Mode (Auto/Manual).

Auditory Feedback: 5V Active Buzzer

This serves as the "Critical Fault Alarm." It triggers for high-priority events such as **Dry Run Detection** or **System Error**, ensuring the operator is alerted even when not viewing the display.

Tactile Inputs (Industrial Push Buttons)

Toggles between *Fully Automated, Time bounded* and *Semi-Automated* states.

- **Manual Override:** Allows for emergency pump activation/deactivation.
- **Supply Acknowledge:** Provides the "Municipal Water Active" signal to the controller.
- **Timer/set:** Provides timer options for the user.

Enclosure: Custom 3D-Printed Housing

Designed using CAD software to consolidate all components into a wall-mountable, ergonomic control unit.

BareBone-OS Protocol v4.0

Architecture: Star Topology (Master + 2 Slaves) **Medium:** 433MHz RF (FSK Modulation) **Bitrate:** 9600 bps

System Architecture and Medium

The BareBone-OS network operates on a high-speed **Star Topology**. By utilizing **FSK Modulation** at **9600 bps**, the system achieves a 4.8x increase in data throughput compared to the prototype. This increased bandwidth is critical for supporting the new **Priority Packet Queue** and **Double-Tap Fault Reporting**.

The network is governed by a **Collision Avoidance Strategy** using staggered response windows:

- **Master (0x10):** 250ms backoff (Highest Priority)
- **SC2 (0x02):** 500ms backoff (High Priority - Pump/Sump)
- **SC1 (0x01):** 1300ms backoff (Normal Priority - Overhead)

Packet Queuing

To ensure reliable and prioritized packet delivery across the WACP v4.0 network, each station implements an 8-slot outbound packet queue. Packets are inserted into the queue based on a four-tier priority scheme: Priority 1 (Critical) is reserved for fault notifications and amperage reports, Priority 2 (Important) handles pump state changes and timer commands, Priority 3 (Normal) is assigned to periodic level reports and ping responses, and Priority 4 (Low) is used for acknowledgement packets. The queue is sorted at insertion time, ensuring that higher priority packets are always transmitted first regardless of their order of arrival. Each queued entry tracks its retry count and acknowledgement status independently. In the event that the queue reaches full capacity, the lowest priority packet is evicted to accommodate an incoming packet of higher priority — a condition that is considered unlikely under normal operating conditions given the low message frequency of the network. Packets remain in the queue until a valid acknowledgement is received from the destination station, at which point they are removed and the next entry is processed. If no acknowledgement is received within the station's defined backoff period, the packet is retransmitted up to a maximum of three attempts before being discarded gracefully. This queuing mechanism eliminates the risk of critical fault data being lost due to simultaneous transmissions or transient channel interference.

Device Addressing and Identification

Each node is assigned a unique 8-bit hexadecimal address. In the v4.0 architecture, the roles are defined as follows:

- **Master Control Unit (0x10):** The central coordinator. Manages the 9-step timer, HMI, and global safety locks. It initiates pings and processes telemetry from both satellites.
- **Overhead Station (SC1 - 0x01):** It now monitors the **Roof Tank**. It uses DHT11-calibrated ultrasonic sensing to report levels and prevents overflows by enforcing the **80% TANK_HIGH** limit.
- **Sump & Pump Station (SC2 - 0x02):** The system's "Muscle." It manages **P1 (Municipal)** and **P2 (Transfer)**. It handles the dual ACS712 multiplexing and manages the **isLocked** safety states.

Device	Address (HEX)	Description
MASTER	0x10	Central Control Unit (NodeMCU + 4 Buttons)
SC1	0x01	Sump Station (Monitor Only)
SC2	0x02	Roof Station (Dual Pump Control + Monitor)

Packet Structure Standard

All communication across the WACP v4.0 network adheres to a fixed 6-byte packet structure to ensure consistency, simplicity, and efficient use of the 9600bps HC-12 radio channel. The first two bytes constitute the preamble, defined as 0xAA followed by 0x55, which serves as a synchronization signature allowing receiving stations to identify the start of a valid transmission. The third byte specifies the destination address, uniquely identifying the intended recipient station — 0x10 for the Master Control, 0x01 for SC1, and 0x02 for SC2. The fourth byte carries the command identifier, which instructs the receiving station on how to interpret and act upon the packet. The fifth byte is the value payload, carrying data relevant to the command such as a water level percentage, pump state, fault code, or encoded amperage reading. The sixth and final byte is the checksum, computed as the bitwise XOR of the destination, command, and value bytes, providing a lightweight error detection mechanism to reject corrupted or malformed transmissions. Upon reception, each station validates the preamble, destination address, and checksum before processing the packet — any failure at any of these three checks results in the packet being silently discarded and the receive buffer being flushed. At 9600bps, a single 6-byte packet occupies approximately 5 milliseconds of channel time, making the protocol highly efficient for the low-frequency reporting demands of the system.

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
PRE_H	**PRE_L**	**DEST**	**CMD**	**VAL**	**CS**
0xAA	0x55	Target ID	Command	Payload	Checksum

Communication Modalities

The BareBone-OS v4.0 implements a **Priority-Stacked Communication Strategy**. This ensures that while the system maintains a constant "Heartbeat," critical safety data always preempts routine telemetry.

1. Timely Communication (The 60s Heartbeat)

To maintain network vitality without clogging the 433MHz frequency, the system uses a 60-second synchronization cycle:

- **Master Polling:** The Master proactively pings satellites every 60 seconds.
- **Slave Silence:** If a station has not had a reason to transmit (no level change, no pump change) for 60 seconds, it sends a "Keep-Alive" health report.

- **Collision Avoidance:** Because the Master (251ms), SC2 (383ms), and SC1 (521ms) use staggered backoff timers, the heartbeat responses are naturally offset, minimizing the risk of packet loss during the sync window.

2. Hybrid Communication (Calibrated Level Monitoring)

Water level data is transmitted using a delta-based logic to preserve bandwidth:

- **The 5% Rule:** A station only generates a level report if the current water level (compensated by **DHT11 temperature data**) differs by more than **5%** from the last transmitted value.
- **Queue Placement:** These reports are assigned **Priority 3 (NORMAL)**. If a higher-priority packet (like a Pump Command) is in the 8-slot queue, the level report will wait until the channel is clear.

3. Event-Driven Communication (The "Priority Jump")

Critical changes bypass all timers and are pushed to the top of the **8-slot Priority Queue** for immediate broadcast:

- **Pump State Changes (Priority 2):** When a pump toggles ON or OFF, or when a **Timer Preset** is updated/expired, the station transmits immediately.
- **Fault & Amperage "Double-Tap" (Priority 1):** This is the highest priority event in the system.
    1. **Step 1:** The station stops the pump and sends `CMD_FAULT`.
    2. **Step 2:** The station immediately follows with `CMD_AMP_Px` containing the trip-current.
    3. **Step 3:** The Master holds the radio channel open specifically to receive this burst before allowing other traffic.

4. Reliability & The "Stop-and-Wait" Protocol

Regardless of the modality (Heartbeat or Event), every packet requires an **ACK (Acknowledge)**.

- **Retry Logic:** If an ACK is not received within the backoff window, the station keeps the packet in its **Priority Queue** and re-attempts the transmission.
- **Queue Depth:** With **8 slots**, the station can buffer multiple events (e.g., a Level Report + a Fault + a State Change) during a brief interference event without losing data.

Control Logic and Safety Standards

The system operates under a **Distributed Authority** model. While the Master issues "Requests," the satellite stations (specifically SC2) perform a local safety audit before execution to ensure hardware integrity and prevent resource wastage.

The Local Safety Audit (The Triple-Check)

Whenever a manual or automated "START" command is received by SC2, it must pass three consecutive gates:

- **Gate 1: The Lock Check:** `IF (isLocked == TRUE)`. If a previous fault (Dry Run/Overload) hasn't been manually cleared by the user, the command is immediately rejected.
- **Gate 2: The Level Check:** Even if the Master requests a fill, SC2 will ignore the command if the target tank level is **≥80%** (detected via the temperature-calibrated JSN-SR04T). This provides a 20% safety buffer.
- **Gate 3: The Watchdog Check:** For Pump 2 (P2), the command is only executed if the **Disconnect Watchdog** is within its 120s heartbeat window.

Pump Identification & State Mapping

To streamline the **8-slot Priority Queue** and ensure the Master's UI reflects the real-world state, pump statuses are encoded as follows:

Pump	State	Integer Value	Description
P1	**OFF**	0	Relay open, pump idle.
P1	**ON**	1	Relay closed, municipal draw active.
P1	**LOCKED**	4	**(NEW) Fault detected; requires manual reset.**
P2	**OFF**	2	Relay open, transfer idle.
P2	**ON**	3	Relay closed, C1 to C2 transfer active.
P2	**LOCKED**	5	**(NEW) Fault detected; requires manual reset.**

Command Rejection Feedback

If a command is rejected by a local safety override (e.g., trying to start a locked pump), SC2 doesn't just stay silent. It pushes a **Priority 2 (IMPORTANT)** packet back to the Master

containing the current `LOCKED` or `OFF` state. This forces the Master's UI to "snap back" to the correct state, informing the user that the command was not executed.

Command Structure

The command structure for the BareBone-OS Protocol v4.0 is categorized by priority to ensure critical safety data preempts routine telemetry. Priority 1 (Emergency) commands include CMD_FAULT (0x50), which triggers the "Double-Tap" sequence, and the amperage telemetry reports CMD_AMP_P1 (0x40) and CMD_AMP_P2 (0x41), which transmit the trip-current (scaled as VAL × 10). Priority 2 (Action) commands handle the operational logic, featuring CMD_PUMP (0x10) for manual relay control (Value 0-3), CMD_TIMER (0x12) for initiating the 9-step scheduling windows, and CMD_UNLOCK (0x15) for clearing a persistent safety lock. Priority 3 (Telemetry) is reserved for CMD_REPORT_LEVEL (0x30), which carries the temperature-compensated water percentage, and CMD_PING (0x01) for active health polling. Finally, Priority 4 (System) consists of the CMD_ACK (0x02) for packet validation and CMD_RETRY (0x03) for handling checksum failures. This hierarchy ensures that a high-stakes event, such as an overload detected via the 0x40/0x41 commands, jump to the front of the 8-slot queue to be processed within the Master's 250ms backoff window.

Command Name	HEX	Sender	Payload (VAL) Definition
CMD_PING	0x01	Master	0x00 (Empty) - Requests immediate status update.
CMD_ACK	0x02	All	Returns the CMD ID being acknowledged.
CMD_REPORT	0x31	SC1	Water Level % (0-100).
CMD_REPORT	0x32	SC2	Water Level % (0-100).
CMD_PUMP	0x20	Master	0=P1 OFF, 1=P1 ON, 2=P2 OFF, 3=P2 ON.
CMD_PUMP	0x20	SC2	Same as above (Reports state change).
CMD_TIMER	0x21	Master	Duration in Minutes (e.g., 30, 60, 90).
CMD_FAULT	0x30	SC2	See Fault Code Table below.

Fault Code Definitions

The BareBone-OS v4.0 utilizes an expanded 8-bit fault register to distinguish between hydraulic, electrical, and communication failures. By using the **ACS712 via the D7 Multiplexer**, the system can now identify "Zero-Current" states, which point to hardware breakage (blown fuses or broken wires) rather than just dry pipes. Additionally, the new **`isLocked` safety state** is automatically applied to any pump triggering codes `0x01` through `0x06`. Unlike the original design, these faults are transmitted via a "Double-Tap" sequence where the Fault Code is immediately followed by a second packet containing the

exact trip-amperage, allowing the Master to display precise diagnostic data (e.g., "Overload @ 9.2A") to the user.

Fault Code (HEX)	Fault Name	Trigger Condition	System Action
0x01	**P1_DRY**	P1 Amps < 0.5A	Stop P1, Set isLocked, Send Double-Tap
0x02	**P1_OVER**	P1 Amps > 8.0A	Stop P1, Set isLocked, Send Double-Tap
0x03	**P1_OPEN**	P1 Amps == 0.0A	Alert Master, Set isLocked (Electrical Fail)
0x04	**P2_DRY**	P2 Amps < 0.5A	Stop P2, Set isLocked, Send Double-Tap
0x05	**P2_OVER**	P2 Amps > 8.0A	Stop P2, Set isLocked, Send Double-Tap
0x06	**P2_OPEN**	P2 Amps == 0.0A	Alert Master, Set isLocked (Electrical Fail)
0x07	**COMMS_LOST**	No Master Ping > 120s	SC2 Freezes P2, SC1 enters Low Power
0x08	**C1_EMPTY**	Reservoir Level < 20%	Forced Stop P2 (Interlock)

Latency Characterisation

In the happy path scenario where the queue is empty and no collisions occur, a command travels from Master to SC2 and receives confirmation in approximately **42ms** — 16ms for transmission, 5ms for SC2 processing, and 16ms for the return ACK. In the realistic worst case where 2–3 lower priority packets are ahead in the queue and one retry is needed due to a collision, end-to-end latency rises to approximately **1.2 seconds**, which remains well within acceptable response time for a water automation system. The absolute worst case — where both Master's and SC2's queues are fully saturated with 7 competing packets each, every transmission collides and exhausts all 3 retries in both directions, and SC2's ultrasonic sensor is mid-read when the packet arrives — produces a theoretical maximum latency of approximately **6.7 seconds**; however this scenario requires every possible failure condition to occur simultaneously and is considered statistically negligible in a 3-node system operating on a dedicated HC12 frequency. It is important to note that safety-critical events such as fault reports and emergency stops are assigned PRIORITY_CRITICAL and jump immediately to the front of the queue, guaranteeing delivery confirmation in approximately **37ms** regardless of queue state or network conditions.

BareBone OS

BareBone OS is a lightweight, cooperative, real-time operating system (RTOS) designed specifically for embedded resource management. Unlike general-purpose operating

systems (like Windows or Linux) that manage complex files and users, BareBone OS is optimized for **determinism**—ensuring that critical tasks (like stopping a pump during a fault) happen immediately, without delay.

Control Core Architecture

Sequential Polling & Network Synchronization

The Master governs the 433MHz channel through a **Staggered Polling Matrix**. Instead of flooding the network, it uses a time-sliced heartbeat strategy (pinging SC1 at the 1s mark and SC2 at the 31s mark). This intentional 30-second offset ensures the Master maintains a constant "pulse" on both satellite stations without creating packet collisions, providing a stable foundation for the Star Topology.

Multi-Tiered Time Management

The architecture operates across three distinct temporal resolutions:

- **Micro-scale (Real-time):** Instantaneous button debouncing and radio packet handling (6-byte buffer monitoring).
- **Meso-scale (Minutes):** Screensaver timeouts and 30-second countdown updates for the 9-step timer.
- **Macro-scale (Daily): NTP Synchronization** via Wi-Fi to correct internal crystal drift, ensuring the system clock remains accurate for 24-hour scheduling cycles.

Responsive HMI State Machine

The Master implements an **Interrupt-Style UX Logic**. Even though it is busy with network coordination, the analog button handler can "wake" the system from its low-power screensaver state immediately. This ensures zero perceived latency for the operator. The UI is decoupled from the automation; it serves as a window into the **evaluateAutoLogic()** function, which acts as the system's primary decision engine based on the data aggregated from the satellites.

Asynchronous Queue Processing

The Master serves as the network's **Data Clearinghouse**. By placing `processQueue()` at the end of every cycle, the OS ensures that high-priority user commands (like a manual Pump Stop) or automatic logic decisions are never blocked by UI rendering or clock updates. Commands are buffered and released only when the channel is clear and the staggered backoff window allows.

In essence, the Master is a **Telemetry Aggregator and Policy Enforcer**. It takes raw data from the "Eyes" (SC1) and "Muscle" (SC2), validates it against the user's manual settings, and issues synchronized directives to keep the entire reservoir system in equilibrium.

SC1 Core Architecture

1. **Environmentally Compensated Sensing:** The OS acknowledges that ultrasonic distance is a variable of air temperature. By interleaving a **DHT11 thermal check** every 10 minutes, the architecture dynamically recalibrates its "Speed of Sound" constant. This ensures that the **80% TANK_HIGH** threshold is a physical reality rather than a calculated guess, preventing overflows during high-heat or cold-weather shifts.
2. **Synchronous/Asynchronous Hybrid Reporting:** The node operates on a "Speak when spoken to, or when things change" logic. It listens for Master polling (Synchronous) but maintains a local **Comparator Circuit** (Asynchronous) that triggers an immediate broadcast if it detects a level shift greater than 5%. This hybrid approach ensures the Master's UI is always accurate during rapid drainage without spamming the network when the water is static.
3. **Radio-Protocol Deference:** SC1 implements a "Polite Listener" strategy. Before performing the computationally blocking task of reading the 1-wire DHT11 sensor, it checks the **HC-12 buffer status**. If there is incoming data from the Master, the OS defers the environmental sensing. This ensures that the node never goes "blind" to the network while it is busy calculating temperature, maintaining a 100% response rate to Master Pings.

In essence, SC1 is a **Deterministic Observer**. It is stripped of high-power control logic to focus entirely on **Data Integrity and Precision**, ensuring that the "Brain" of the system makes decisions based on the most accurate telemetry possible.

SC2 Core Architecture

Asynchronous Scheduling: The system avoids the "Freeze" typical of simpler controllers. By using internal timestamps instead of delay pauses, the OS can simultaneously track a 30-minute pump timer, monitor real-time amperage, and listen for radio pings without one task slowing down the other.

Channel-Aware Prioritization: The OS is "radio-first." It prioritizes the communication buffer above all else. For instance, it will actually skip a routine temperature reading if the radio channel is busy, ensuring that a "Stop" command from the Master is never missed due to local sensor processing.

Delta-Based Intelligence: To save power and radio bandwidth, the OS is designed to be "quiet." It doesn't report data just for the sake of it; it only triggers a transmission if a significant physical event occurs (like a 5% shift in water level) or if a safety threshold is breached.

Buffered State Management: The logic is decoupled from the transmission. When a fault occurs, the "Brain" of the OS instantly shuts down the hardware and pushes a "Double-Tap" data report into an 8-slot Priority Queue. This ensures that even if the radio environment is noisy, the station won't "forget" to report a fault; it will keep the data in its stack and retry until the Master acknowledges the event.

Future Expansions (Roadmap)

Since BareBone OS is built on a modular architecture using the ESP8266, it is highly scalable. Here are the most logical next steps for v5.0 and beyond:

Water Quality Monitoring

- **Concept:** Adding support for analog sensors on the I2C bus.
- **Feature:** Integrate pH, Turbidity (clarity), or TDS (Total Dissolved Solids) sensors at the Basement Station (SC2).
- **Value:** The system could refuse to pump water up to the roof if the source water is too dirty or acidic, protecting the household's health.

Mesh Networking (Range Extension)

- **Concept:** Move from a "Star Topology" (Master <-> Slave) to a "Mesh Topology."
- **Feature:** If any of the stations is far from the control, the station in range behaves as a repeater.
- **Value:** Infinite range extension for large agricultural farms or tall apartment complexes.

Touch Interface (HMI Upgrade)

- **Concept:** Replace the 4-Button + LCD setup with a Nextion Touch Display or an OLED.
- **Feature:** A drag-and-drop UI for setting timers, viewing historical graphs of water usage, and easier diagnostics.
- **Value:** Modernizes the look and feel for high-end consumer markets.

BareBone OS (v4.0) System Performance Analysis

Date: January 28, 2026 **Architecture:** Star Topology (Central Master, Distributed Slaves) **Protocol:** Hybrid Time-Division / Event-Driven (HTDE) **Medium:** 433MHz ISM Band (ASK Modulation)

Physical Layer Efficiency (RF Link Budget)

To validate the reliability of the HC-12 communication hardware over the target distance of 100 meters, a Friis Transmission Equation analysis was conducted to determine the Free Space Path Loss (FSPL) and resulting link margin under different supply voltage conditions.

Free Space Path Loss Calculation: The theoretical signal degradation over distance is calculated using the logarithmic form of the Friis equation:

$$FSPL(dB) = 20\log10(d) + 20\log10(f) - 27.55$$

Where:

- d = Distance in meters (100m)
- f = Frequency in MHz (433MHz)

Substituting the system parameters:

$$FSPL = 20\log10(100) + 20\log10(433) - 27.55$$

$$FSPL = 40 + 52.73 - 27.55$$

$$FSPL \approx 65.18 \text{ dB}$$

Link Margin Assessment: Given the hardware specifications of the NodeMCU RF interface and transmitting power of 5v.

- **Transmitter Power (PTX):** +20 dBm
- **Receiver Sensitivity (RX):** -116 dBm (Standard superheterodyne floor)

The Link Margin is calculated as:

$$Margin = (PTX - FSPL) - RX$$

Margin = (20−65.18) − (−116)

Margin = +70.82 dB

This represents extremely robust communication capability with strong tolerance to walls, floor separation, and environmental interference.

Link Margin 2nd Assessment: Given the hardware specifications of the NodeMCU RF interface and transmitting power of 3.3v.

- **Transmitter Power (PTX):** +13 dBm
- **Receiver Sensitivity (RX):** -116 dBm (reduced output mode)

The Link Margin is calculated as:

Margin = (PTX−FSPL) − RX

Margin = (13−65.18) − (−116)

Margin = +63.82 dB

Even with reduced supply voltage, the communication link maintains a very large safety margin.

Conclusion: Both operating conditions provide exceptionally high link margins (>60 dB). In practical RF systems, a margin of 10–20 dB is typically sufficient for reliable non-line-of-sight communication. The HC-12 therefore offers significantly improved reliability compared to basic ASK modules and ensures stable operation across multi-floor structural barriers and indoor environments.

The 5V configuration provides maximum robustness, while the 3.3V configuration still maintains more than adequate performance for the water automation system.

Protocol Efficiency & Bandwidth Utilization

The v4.0 protocol utilizes a hybrid heartbeat mechanism to maintain synchronization without saturating the channel.

Duty Cycle Calculation:

- **Bitrate:** 9600 bps
- **Payload:** 6 Bytes (approx. 150 bits OTA including preamble/encoding)
- **Transmission Time (TTX):** 150/9600=0.016s

- **Transaction Time (TTotal):** 0.016s(Ping) + 0.016s(Ack) + 0.05s(Process) = 0.082s

With a polling interval of 60 seconds for two devices, the total Active Time per minute is 0.082 seconds.

Channel Efficiency = Idle Time/Total Time × 100

Efficiency = 59.918/60 × 100 = 99.86%

Conclusion: The HC-12 implementation reduces packet transmission time by more than 50% compared to the previous 2000 bps implementation. The channel remains idle for approximately **99.86%** of operational time, representing an extremely low duty cycle.

This results in:

- Lower channel congestion
- Minimal probability of packet collisions
- Improved system responsiveness
- Better coexistence with other 433 MHz devices

The higher data rate therefore improves both bandwidth utilization and communication efficiency while maintaining reliable synchronization between system nodes.

Collision Resilience: The "Asymmetric Backoff" Strategy

A critical vulnerability in broadcast networks is the "Deadlock" scenario, where two stations transmit simultaneously, collide, wait the same amount of time, and collide again.

To mitigate this, the BareBone OS implements a deterministic **Asymmetric Backoff Algorithm**:

- **SC2 (Pump Station):** Retry Interval = **383ms**
- **SC1 (Roof Station):** Retry Interval = **521ms**

"The Storm Scenario": Under simulated conditions of heavy traffic (Queue Saturation) combined with stochastic packet loss (e.g., Thunderstorm interference):

1. **Initial State:** Master Heartbeat triggers simultaneous response from SC1 and SC2. **(Collision occurs)**.
2. **Resolution Phase:**
    - SC2 wakes up at T+383ms. The channel is clear. It transmits successfully.
    - SC1 wakes up at T+521ms. The channel is clear. It transmits successfully.

3. **Zipper Effect:** This timing difference forces the stations into a serialized queue. SC2 effectively "takes over" priority, clearing its critical pump commands before SC1 attempts to report water levels.

Conclusion: The probability of a sustained deadlock loop is reduced to almost **0%**. The system signals will collide once per about **13hrs** that leaves enough time to process the deadlock.

Final Performance Analysis Matrix

Metric	Measured/Calc Value	Grade	Engineering Notes
Link Budget	(+63.82 dB)	**A+**	Signal strength is sufficient to penetrate concrete walls or 200m+ LOS.
Bandwidth Efficiency	99.63% Idle	**A+**	"Military Grade" silence. Leaves ample room for future expansion or repeaters.
Latency (Normal)	< 200 ms	**A**	Human-imperceptible delay for button presses.
Latency (Worst Case)	1.2 Seconds	**A**	Collision recovery is deterministic and rapid, enabling near real-time pump control and fluid level management.
Deadlock Risk	9 collison/min	**A+**	Asymmetric timers prevent recursive loops.
Throughput	~450 Packets/10s	**High**	Considering about 10 Pck as realistic, utilisation is around 2.2%

System Testing and Experimental Validation

To ensure the reliability and safety of the BareBone OS architecture, a series of controlled experiments were designed to validate the physical layer, sensor accuracy, and logic safety interlocks.

Experiment A: Communication Protocol Robustness

Objective: To determine the maximum effective range of the 433MHz RF link, quantify packet loss rates under load, and verify the "Asymmetric Backoff" collision handling mechanism.

A.1 Range and Packet Loss Test

- **Apparatus:** Master Unit (Stationary), Satellite Station (Mobile), Distance Measure/ Pre determined distance.
- **Methodology:**
    1. Place the Master Unit in a fixed location (central hub).
    2. Move the Satellite Station away in 10-meter increments.
    3. At each increment, send 20 "PING" packets from the Master.
    4. Record the number of successful "ACK" responses.
    5. Continue until packet loss exceeds 50% (Critical Failure Point).

Data Table:

Distance (m)	Packets Sent	ACKs Received	Packet Loss (%)	Link Quality
10m	20		…	…
30m	20	…	…	…
50m	20	...	...	……
100m	20	...	...	……

A.2 Collision Handling Verification

- **Objective:** To verify that SC2 (Pump Station) takes priority over SC1 (Sump) during simultaneous transmission.
- **Methodology:**
    1. Force a reset on both SC1 and SC2 simultaneously so their timers align.
    2. Monitor the Data received at the Master Unit.
    3. Observe the arrival time of the packets.
    4. **Success Criteria:** The Master must receive the SC2 packet *before* the SC1 packet, and both must arrive within 3 seconds without corruption.

Experiment B: Water Level Sensor Calibration

Objective: To verify the accuracy of the ultrasonic measurements and test the software's ability to filter out noise caused by water turbulence (e.g., during tank filling).

B.1 Static Accuracy Test

- **Apparatus:** Bucket/Tank, Meter Ruler, SC1/SC2 Unit.
- **Methodology:**

1. Fill the tank to specific levels (Empty, 25%, 50%, 75%, 100%).
2. Measure the actual water level (distance from sensor) using the meter ruler.
3. Record the displayed value on the Master Unit.
4. Calculate the deviation (Error %).

Data Table:

Physical Level (cm)	Sensor Reading (cm)	Displayed %	Error Margin (cm)
100 cm	…	…	…
50 cm	…	…	…
10 cm	…	…	…

B.2 Turbulence & Hysteresis Test

- **Methodology:**
  1. Fill the tank to 80%.
  2. Manually disturb the water surface to create waves (simulating pump inflow).
  3. Observe the Master Display for 60 seconds.
  4. **Success Criteria:** The display percentage should remain stable (e.g., sticking at 80-81%) rather than fluctuating wildly (e.g., jumping 70% -> 90% -> 75%), proving the software filter is working.

Experiment C: Current Sensor & Fault Analysis

Objective: To validate the accuracy of the ACS712 current sensor and the response time of the system in detecting pump faults (Dry Run / Overload).

C.1 Load Accuracy Test

- **Apparatus:** Multimeter (Clamp Meter), Pump/Load, ACS712 Module.
- **Methodology:**
  1. Connect the pump load.
  2. Measure the actual current draw using the Clamp Meter.
  3. Record the current value calculated by the BareBone OS.
  4. **Formula:** Accuracy = (1 − Measured/Measured−Sensor|) × 100

C.2 Fault Trigger Latency

- **Methodology:**
    1. **Simulate Dry Run:** Disconnect the pump intake (No water).
    2. Send the "START" command.
    3. Measure the time taken for the Master to receive the "DRY RUN" error and lock the screen.
    4. **Simulate Overload:** Stalled rotor simulation (or high resistive load > 8A).
    5. Measure the time taken for the system to cut power and report "OVERLOAD."

Data Table:

Fault Type	Threshold Set	Actual Load	Trip Time (sec)
Dry Run	…	…	…
Overload	…	…	…

Experiment D: System Logic & Safety Interlocks

Objective: To validate the "Fail-Safe" logic that prevents manual user errors, specifically attempting to pump water into an already full tank.

- Apparatus: Complete System Setup.
- Methodology:
    1. Manipulate the Roof Station (SC2) sensor to simulate a FULL TANK (Distance < 10cm).
    2. Ensure the Master Display shows "ROOF: 100%".
    3. Press the BTN_START (Manual Start) on the Master Unit.
    4. Observation: Does the pump start?
    5. Expected Result: The system should reject the command, keep the pump OFF, and potentially display a status message (e.g., "TANK FULL").

Test Case	Initial State	Action	System Response	Result (Pass/Fail)
Logic Lock	Roof = 100%	Press Start	…	**…**
Recovery	Roof = 80%	Press Start	…	**…**